

Trench photo deficiency report

Demo dataset · Generated 17 May 2026, 01:52

This report flags the sections of your fiber-trench run that didn't clear every quality check, so you know exactly what to fix before the job can close out. Bring it to the site, work through the flagged sections one by one, and re-shoot or correct whatever each card asks for. Nothing in here requires office action — every item is something the crew can resolve in the field.

At a glance

20	5	5	10
SECTIONS SCORED	PASSING	WARNINGS	NEEDS REVIEW

Of the 20 sections reviewed, 5 passed clean, 5 came back with warnings worth watching, and 10 are hard fails that need action before sign-off. The story is almost entirely one thing: half the sections have no photos attached at all, which by itself accounts for all 10 red verdicts. Beyond that, five individual sections each carry their own separate problem — a 31-metre gap between shot points, a location that plots 1.7 km away from where it should be, a segment where warning tape wasn't visible during the tape-laying phase, one section where a photo was reused from somewhere else, and one where the only photos on file were pulled for privacy or irrelevance and nothing usable was left. Your immediate next step is to take the crew out and photograph every unshot section, then circle back to address those five one-off issues with the specific fixes called out on each card.

Run details

Identifies the pipeline run that produced this report: which AI model the per-photo checks used, the run cost, the total trench length scored, and the thresholds the verdicts use.

AI model	claude-sonnet-4-6
Run source	Demo dataset
Generated	17 May 2026, 01:52
Photos processed	23
AI cost	\$0.11
Sections scored	20
Trench scored	851 m (245 m passing · 28.8%)
Coverage rule	1 compliant photo per 5 m of trench
Off-route cutoff	Photos > 75 m from any trench are ignored as evidence

Photo intake

These numbers describe the photos that came in for this run. "Passed every per-photo check" is the pool of usable photos that actually count toward the one-photo-every-five-metres coverage rule — everything else either got rejected

(faces visible, off-topic, unreadable) or never arrived in the first place.

Photos uploaded	23
Passed every per-photo check	21
Duplicates caught (same photo re-submitted)	1
Flagged for re-shoot (faces / licence plates)	1
Not usable for QC (portrait, off-topic, unreadable)	1
GPS-vs-address mismatch (photo not where it claims)	1
Image possibly re-saved (low-confidence signal)	1

The rest of this document is organised by FCP route: sections that share the same 'no photos' problem are collapsed into a single summary block per route to keep things easy to scan, while sections with a unique issue each get their own full card spelling out exactly what went wrong and what to do. Passing sections and the complete list of eight quality checks are in the appendix at the back.

Sections needing attention

Each card below names a section that did not pass every check, the action the crew should take, and (when the data is available) every photo that contributed to the verdict with the AI's per-check answer.

FCP F001 · 4 sections flagged

Section S005 · F001

5 Mockstrasse, Synthville

Warning

Length 50 m · 1 of 2 photos passed the per-photo checks · Biggest uncovered stretch 12 m

DO NEXT

Re-shoot 1 photo(s) without faces or licence plates in frame. Re-shoot 1 photo(s) showing the actual trench, not other subjects. Re-shoot enough photos to cover this section after the excluded ones.

EVIDENCE — PHOTOS ON THIS SECTION

fx_1_img_20240911_wa0091.jpg · other stage · excluded — off topic

Addr 5 Mockstrasse, Synthville · Stamped 26.08.2024 14:30 · Pos t=0.10, 1.2 m off centreline
warning tape ✗ · sand bedding ✗ · side view ✗ · depth ref ✗ · duct ✗ · ends sealed n/a · personal data ✓
"Dumpster / staging area; not a trench photo."

fx_1_img_20240814_wa0046.jpg · paper-label stage · excluded — personal data

Addr 5 Mockstrasse, Synthville · Stamped 26.08.2024 14:30 · Label F001-R001-15-or · Pos t=0.50, 1.2 m off centreline
warning tape ✗ · sand bedding ✗ · side view ✗ · depth ref ✗ · duct ✗ · ends sealed n/a · personal data ✗
"Paper label close-up; worker's face partially visible."

Section S006 · F001

22 Beispielweg, Synthville

Warning

Length 50 m · 1 of 2 photos passed the per-photo checks · Biggest uncovered stretch 12 m

DO NEXT

Re-shoot fresh photos for this section — duplicates of older submissions were detected.

EVIDENCE — PHOTOS ON THIS SECTION

fx_1_img_20240814_wa0048.jpg · duct-laying stage · counted toward coverage

Addr 22 Beispielweg, Synthville · Stamped 26.08.2024 14:30 · Label F001-R001-16-or · Pos t=0.50, 1.2 m off centreline
warning tape ✓ · sand bedding ✓ · side view ✓ · depth ref ✗ · duct ✓ · ends sealed ✓ · personal data ✓
"Same image was submitted to job #DEMO-A1 and #DEMO-A2."

fx_dup_1_img_20240816_wa0041.jpg · duct-laying stage · inherited — duplicate

Addr 22 Beispielweg, Synthville · Stamped 26.08.2024 14:30 · Label F001-R001-16-or · Pos t=0.50, 1.2 m off centreline
warning tape ✓ · sand bedding ✓ · side view ✓ · depth ref ✗ · duct ✓ · ends sealed ✓ · personal data ✓
"Same image was submitted to job #DEMO-A1 and #DEMO-A2."

Section S008 · F001

78 Wrongtown-Strasse, OffCluster

Warning

Length **50 m** · 0 of 1 photo passed the per-photo checks · Biggest uncovered stretch **12 m**

DO NEXT

Re-take photos here — the GPS is 1.7 km away from the printed address; the photo is not where it claims.

EVIDENCE — PHOTOS ON THIS SECTION

fx_1_img_20240822_wa0019.jpg · duct-laying stage · excluded — GPS / address disagree

Addr 78 Wrongtown-Strasse, OffCluster · Stamped 26.08.2024 14:30 · Label F002-R001-12-or · Pos t=0.50, 1.2 m off centreline · GPS 46°33'18.0"N 14°17'21.0"E
warning tape ✓ · sand bedding ✓ · side view ✓ · depth ref ✗ · duct ✓ · ends sealed ✓ · personal data ✓
"Coordinates and printed address disagree by 1.7 km — flagged off-cluster."

Section S004 · F001

3 Mockstrasse, Synthville

Warning

Length **47 m** · 1 of 2 photos passed the per-photo checks · Biggest uncovered stretch **31 m**

DO NEXT

Re-shoot one photo every 5 m between meter 2 and meter 33 (currently a 31 m gap).

EVIDENCE — PHOTOS ON THIS SECTION

fx_1_img_20240813_wa0036.jpg · duct-laying stage · counted toward coverage

Addr 3 Mockstrasse, Synthville · Stamped 26.08.2024 14:30 · Label F001-R001-14-or · Pos t=0.05, 1.2 m off centreline
warning tape ✓ · sand bedding ✓ · side view ✓ · depth ref ✗ · duct ✓ · ends sealed ✓ · personal data ✓

fx_1_img_20240814_wa0045.jpg · sand-bedding stage · counted toward coverage

Addr 3 Mockstrasse, Synthville · Stamped 26.08.2024 14:30 · Label F001-R001-14-or · Pos t=0.70, 1.2 m off centreline
warning tape ✓ · sand bedding ✓ · side view ✓ · depth ref ✗ · duct ✓ · ends sealed ✓ · personal data ✓

FCP F002 · 11 sections flagged

Section S015 · F002

1 Mockstrasse, Synthville

Warning

Length **43 m** · 0 of 1 photo passed the per-photo checks · Biggest uncovered stretch **12 m**

DO NEXT

Re-shoot one photo showing the warning tape, during the tape-laying stage.

EVIDENCE — PHOTOS ON THIS SECTION

fx_img_20240723_wa0028_■■■■■■.jpg · tape-laying stage · excluded — warning tape not visible

Addr 1 Mockstrasse, Synthville · Stamped 26.08.2024 14:30 · Label F002-R003-11-or · Pos t=0.50, 1.2 m off centreline
warning tape ✗ · sand bedding ✓ · side view ✓ · depth ref ✗ · duct ✓ · ends sealed ✓ · personal data ✓
"Tape phase but no warning tape visible. Likely missed by contractor."

10 sections with no photos submitted · 367 m of trench, FCP F002**DO NEXT**

Re-shoot 4–6 photos along each section below, one for each work stage (open trench, sand bedding, cable laid, warning tape).

S009 · 43 m

S010 · 38 m

S012 · 32 m

S013 · 36 m

S014 · 45 m

S016 · 43 m

S017 · 36 m

S018 · 32 m

S019 · 34 m

S020 · 29 m

Appendix

Everything below is the audit trail behind the verdicts on the previous pages — the route-level totals, the passing sections, every photo the pipeline saw, the duplicates and GPS mismatches it caught, and the rules that produced these answers.

Route summary (per FCP)

Each FCP is one project-zone route. The table below totals every section in the route by verdict, the trench length they represent, the length share already passing, and the number of photos that landed inside the route.

FCP	Sections	G / Y / R	Trench length	Passing length	Photos
F001	8	4 / 4 / 0	405 m	209 m (52%)	20
F002	12	1 / 1 / 10	446 m	36 m (8%)	4

Passing sections

These sections passed every check — no action needed. Listed for completeness so nothing is left unmentioned.

FCP F001 · S001, S002, S003, S007

FCP F002 · S011

Photo audit log

Every photo processed in this run, with the segment it landed on, the work phase it documents, and how it contributed to the verdict. "Counted" = the photo passed every per-photo check and was used toward the coverage rule. Anything else is excluded with the reason shown.

Photo	Segment	Phase	Status	Address overlay	Cost
fx_1_img_20240731_wa0029_jpg	S001	excavation	counted toward coverage	12 Demo-Straße, Synthville	\$0.0046
fx_1_img_20240731_wa0030_jpg	S001	duct-laying	counted toward coverage	12 Demo-Straße, Synthville	\$0.0046
fx_1_img_20240809_wa0025_jpg	S001	sand-bedding	counted toward coverage	12 Demo-Straße, Synthville	\$0.0046
fx_1_img_20240809_wa0038_jpg	S002	duct-laying	counted toward coverage	14 Demo-Straße, Synthville	\$0.0046
fx_1_img_20240812_wa0033_jpg	S002	sand-bedding	counted toward coverage	14 Demo-Straße, Synthville	\$0.0046
fx_1_img_20240812_wa0034_jpg	S002	tape-laying	counted toward coverage	14 Demo-Straße, Synthville	\$0.0046
fx_1_img_20240813_wa0028_jpg	S003	excavation	counted toward coverage	1 Mockstrasse, Synthville	\$0.0046
fx_1_img_20240813_wa0031_jpg	S003	duct-laying	counted toward coverage	1 Mockstrasse, Synthville	\$0.0046
fx_1_img_20240813_wa0034_jpg	S003	sand-bedding	counted toward coverage	1 Mockstrasse, Synthville	\$0.0046
fx_1_img_20240813_wa0036_jpg	S004	duct-laying	counted toward coverage	3 Mockstrasse, Synthville	\$0.0046
fx_1_img_20240814_wa0045_jpg	S004	sand-bedding	counted toward coverage	3 Mockstrasse, Synthville	\$0.0046
fx_1_img_20240814_wa0046_jpg	S005	paper-label	excluded — personal data	5 Mockstrasse, Synthville	\$0.0046
fx_1_img_20240814_wa0048_jpg	S006	duct-laying	counted toward coverage	22 Beispielweg, Synthville	\$0.0046
fx_1_img_20240816_wa0045_jpg	S007	duct-laying	counted toward coverage	12 Demo-Straße, Synthville	\$0.0046
fx_1_img_20240822_wa0010_jpg	S007	sand-bedding	counted toward coverage	12 Demo-Straße, Synthville	\$0.0046
fx_1_img_20240822_wa0011_jpg	S007	tape-laying	counted toward coverage	12 Demo-Straße, Synthville	\$0.0046
fx_1_img_20240822_wa0019_jpg	S008	duct-laying	excluded — GPS / address disagree	78 Wrontown-Strasse, OffCluster	\$0.0046
fx_1_img_20240911_wa0030_jpg	S011	duct-laying	counted toward coverage	14 Demo-Straße, Synthville	\$0.0046
fx_1_img_20240911_wa0035_jpg	S011	sand-bedding	counted toward coverage	14 Demo-Straße, Synthville	\$0.0046
fx_1_img_20240911_wa0091_jpg	S005	other	excluded — off topic	5 Mockstrasse, Synthville	\$0.0046
fx_1_img_20240911_wa0092_jpg	S007	duct-laying	counted toward coverage	12 Demo-Straße, Synthville	\$0.0046
fx_1_timephoto...240807_182537_jpg	S011	tape-laying	counted toward coverage	14 Demo-Straße, Synthville	\$0.0046

Photo	Segment	Phase	Status	Address overlay	Cost
fx_dup_1_img_20240816_wa0041.jpg	S006	duct-laying	inherited — duplicate	22 Beispielweg, Synthville	\$0.0046
fx_img_2024072...a0028_■■■■.jpg	S015	tape-laying	excluded — warning tape not visible	1 Mockstrasse, Synthville	\$0.0046

Duplicates detected

Photos with identical perceptual fingerprints are grouped into a cluster. One photo from the cluster is treated as the representative and reviewed by the AI; the rest are flagged as duplicates of that submission, so a contractor can't fill a section by re-uploading the same good photo across jobs.

Cluster	Representative photo	Inherited duplicates	Segments touched
1	fx_1_img_20240814_wa0048.jpg	fx_dup_1_img_20240816_wa0041.jpg	S006

GPS-vs-address mismatches

Photos whose printed-overlay address and on-photo GPS coordinates point to different places, or whose GPS lands outside any project zone. These are the photos most worth a human eyeball — accidents look the same as forgeries until someone checks the site.

Photo	Segment	Printed address	On-photo GPS	Snap dist.	Note
fx_1_img_20240822_wa0019.jpg	S008	78 Wrongtown-Strasse, OffCluster	46°33'18.0"N 14°17'21.0"E	1.2 m	Coordinates and printed address disagree by 1.7 km — flagged off-cluster.

Photos flagged for personal data

Faces and licence plates were visible in these photos. They are excluded from coverage scoring and queued for re-shoot without the personal data in frame (GDPR / NIS2 obligations).

Photo	Segment	Phase	AI note
fx_1_img_20240814_wa0046.jpg	S005	paper-label	Paper label close-up; worker's face partially visible.

Photos not used for QC

These photos arrived in the upload but couldn't be used as trench evidence — the AI labelled them off-topic, a portrait, or unreadable. They do not count for or against any section's verdict; they're listed here so nothing is silently dropped.

Photo	Reason	Phase guess	AI note
fx_1_img_20240911_wa0091.jpg	off topic	other	Dumpster / staging area; not a trench photo.

ELA tamper hints

ELA (Error Level Analysis) compares each photo against a freshly re-saved copy of itself. Regions that stand out point to recent edits or re-compression. It is a soft signal — these photos still feed the coverage check, but a reviewer may want to look again.

Photo	Segment	ELA score	AI note
fx_1_img_20240911_wa0092_jpg	S007	0.18	Visually plausible but ELA suggests re-save / re-compression.

Methodology — how a verdict is computed

Each trench segment is scored independently. The pipeline first filters the photos snapped to the segment, then walks the compliant ones from one end to the other looking for gaps.

1. Per-photo filter

A photo counts as evidence only if the AI marked it **scorable** (not portrait, off-topic, paper-label, or unreadable), did **not** see personal data, its printed address agreed with its on-photo GPS, and its snap to the nearest trench is within **75 m**. Duplicates (same perceptual fingerprint) only contribute once per cluster — re-uploading a good photo across jobs does not fill a section.

2. Phase-specific visual checks

The AI assigns each photo a work stage. Only the checks relevant to that stage are required:

Stage	Required checks (each must be "yes")	Counted as evidence?
excavation	side view present	yes
depth-measuring	depth reference visible, side view present	yes
duct-laying	duct visible	yes
sand-bedding	sand bedding visible, duct visible	yes
tape-laying	warning tape visible	yes
back-filled	(no specific check — photo documents the state)	yes
restored	(no specific check)	yes
paper-label	(documents which FCP, not the trench itself)	no
staging	(no trench in frame)	no
other	(unrecognised stage)	no

3. Coverage and verdict

GREEN — the biggest gap between compliant photos is ≤ 5 m, the first compliant photo is within **5 m** of the start, and the last is within **5 m** of the end.

RED — zero compliant photos on the section, or fewer than **1 compliant photo per 10 m** of trench.

YELLOW — anything in between: photos exist but coverage has a gap, or one or more failed a check.

4. Confidence and limits

Each visual check is answered by Claude (an AI model — the version in use is named on the cover). Models can be wrong. The dataset shipped with 219 photos hand-labelled for the depth-measurement and duct-laying stages; the live run's agreement rate against those labels is the one number we use to gauge per-photo accuracy. The geo check (printed address vs on-photo GPS) is deterministic — no model involved. The depth check is intentionally limited to "is a depth reference visible / readable"; we do not yet read the number off the ruler.

Severity colour codes

Needs review Not enough good photos to confirm the work is OK. A reviewer should look at this first.

Warning Some checks failed or coverage has a gap, but the section is mostly documented.

Passing Every check passed and coverage meets the one-photo-per-five-metres rule. No action needed.

What the tool checks on every photo

Six of these are answered by the AI model; the last two are answered by direct comparisons that don't need a model. The phase a photo is in determines which of the six are required (see the methodology section above).

1. **Warning tape visible.** Orange tape on top, so a future digger knows the cable is below.
2. **Sand bedding visible.** Cable sits on sand, not bare dirt — protects against damage.
3. **Side view of trench.** You can only judge depth from a side angle, not a top-down shot.
4. **Depth reference visible.** A ruler or measuring rod in the frame, so depth can be read.
5. **Cable ends sealed.** White end-caps on the duct bundle — keeps water out.
6. **No personal data.** Faces and licence plates are flagged for retake (GDPR).
7. **Not a duplicate photo.** Image fingerprint compared against every other photo — copies caught.
8. **Printed address matches GPS.** The address stamp on the photo agrees with its GPS coordinates.